

Little Caesars

Formulas used in the app

4 October 2026. All times are Asia/Riyadh (UTC+3, no daylight saving).

This note lists every formula the dashboard uses: energy, cost, AC savings, customers, and table occupancy. A number on the screen is one of these, rounded as described at the end.

1. The clocks

Energy day. 06:00 to 06:00 the next morning. A reading before 06:00 belongs to the previous energy day. The energy day that is still open is left out of closed totals. On the afternoon of 4 October 2026, the latest closed energy day is 3 October: 06:00 on 3 October through 06:00 on 4 October.

Customer shift. 10:00 to 05:00 the next morning. Hours 00:00-04:59 belong to the shift that opened the day before. Hours 05:00-09:59 are outside the shift.

Calendar day. Midnight to the next midnight.

2. How a kWh number is made

Each meter stores a running total. Use is the difference between two readings of the same meter.

device kWh = next reading - this reading

The next reading must be the very next slot: the next energy day, the next hour, or the next 15 minutes. A missing slot stays blank. It is not filled with zero. If the next reading is lower than this one, the row stays blank. A result is never stored as a negative.

day total = Lights + AC + Oven + Freezer + Chiller

A load is the sum of the devices in that load. Lights, AC, oven, freezer, and chiller are the five loads.

3. Home energy card

Latest closed day

The last energy day whose date is before today's energy date. The five tiles under the total are that same day, not the month.

Worked example, closed day 3 October 2026. The tiles are Lights 50.1, AC 82.4, Oven 147.3, Freezer 32.5, Chiller 21.4. Added together that is 333.7 kWh. The card shows 333.8 kWh because the total and each tile are rounded to one decimal on their own.

Last 7 days

The seven closed days that end on the latest closed day. For 3 October that is 27 September through 3 October.

change kWh = last 7 days - the 7 days before that

change % = (last 7 days - previous 7 days) / previous 7 days x 100

2,440.2 kWh minus 2,873.0 kWh is -432.8 kWh. The percent is -15%, shown as Reduction -15%. This card uses those rolling seven days. It does not use the Saturday-to-Friday week.

This month

Closed days from the 1st through the latest closed day. The open day is not included. "Vs last month" is the same day numbers in the previous month, not the whole previous month.

change kWh = this month so far - the same days last month

change % = (this month so far - same days last month) / same days last month x 100

1-3 October is 1,103.0 kWh. 1-3 September is 2,938.6 kWh. The difference is -1,835.6 kWh, which is -62%.

The word Reduction is used when the percent is negative. Increase is used when it is positive. The minus sign stays on the percent. If there is no previous period, the card asks for more days. If the previous value is 0 and the current value is 0, the change is 0%. If the previous value is 0 and the current value is not 0, the change is 100%.

4. Other energy comparisons

The chain board uses the same closed-day totals. Some ranges also add live use since 06:00 today.

projected month = (month so far / closed days) x days in the month

The Saudi week starts on Saturday. Week to date on the chain board is Saturday through the latest closed day, compared with the same weekdays one week earlier. The home card's "last 7 days" is a different window.

Same month last year is day 1 through the same day number, one year earlier.

saved kWh = average of (last month same days - this month) and (last year same days - this month)

saved SAR = cost(this month + saved kWh) - cost(this month)

5. Report tables

6:00 AM report. One row per device per energy day. Consumption is the next 06:00 meter minus this 06:00 meter.

Midnight report. Calendar midnight to midnight. Consumption is the next midnight meter minus this midnight meter.

Hourly chart. Each closed hour is the next hour's meter minus this hour's meter, summed for the devices you selected. The current hour is live use since that hour started. Future hours are empty.

AC 15-minute log. Each slot is the next 15-minute meter minus this meter. The quarter that is still open is left out. A missed quarter is blank.

6. Live power and current

live kWh since 06:00 = max(0, meter now - meter at 06:00)

If the 06:00 reading is missing, live use is 0.

AC current. A negative amp reading is stored and shown as a positive number. -10 A is shown as 10 A. The AC total current is the sum of the absolute value of each channel.

AC power. A negative watt reading is shown as 0 W. It is not flipped to a positive number.

7. Electricity cost

Inside one month, the first 6,000 kWh cost 0.22 SAR each. Every kWh after that costs 0.32 SAR. Exactly 6,000 kWh is still entirely at 0.22.

$$\text{cost} = \min(\text{kWh}, 6000) \times 0.22 + \max(0, \text{kWh} - 6000) \times 0.32$$

$$\text{cost of a slice} = \text{cost}(\text{kWh already counted} + \text{slice}) - \text{cost}(\text{kWh already counted})$$

Today's cost is the slice after the closed days already counted this month. A finished month is billed on its own total. The open month on the board includes live use since 06:00.

8. AC automation savings

The green line on the hourly chart. It is shown for all devices, or for AC. Off is not counted as saving. Each AC is calculated on its own, then the hour is the sum.

The occupied target is the lowest target temperature that unit has used while the mode is cool or heat.

A 15-minute slot is a normal cooling slot when the mode is cool or heat and the target is within 0.4 C of that occupied target.

A slot is a saving slot when either of these is true:

The mode is fan only.

The mode is cool or heat, and the target is more than 0.4 C above the occupied target. Example: occupied target 20 C, automation target 26 C.

$$\text{cooling rate} = \text{average of (kWh used in the slot} / 0.25) \text{ over the previous 2 hours of normal slots}$$

0.25 hours is one 15-minute slot, because $15 / 60 = 0.25$. Dividing by 0.25 turns the slot's kWh into kWh per hour.

$$\text{saved kWh in the slot} = \max(0, \text{cooling rate} \times 0.25 - \text{kWh used})$$

The green point for an hour is the sum of those slot savings in that hour. Nothing is counted until a normal-target cooling rate exists. A slot with no mode is neither a saving slot nor a baseline slot.

9. Customers

The camera service stores hours in UTC. The app adds 3 hours so the hour matches Riyadh.

$$\text{shift customers} = \text{entries from 10:00 to 23:59} + \text{entries from 00:00 to 04:59 the next morning}$$

6:00 AM report, for a range of closed energy days:

$$\text{customers} = \text{calendar entries from the start date through the end date}$$

$$- \text{entries from 00:00 to 05:59 on the start date}$$

$$+ \text{entries from 00:00 to 05:59 on the morning after the end date}$$

If the last energy day is still open and the clock is before 06:00, add this morning's 00:00-05:59 entries instead of the next morning. The result is never negative.

Midnight report. Calendar entries from the start date through the end date. No hour is added or removed at the edges.

On the business report, a range longer than 31 days uses calendar-day customer counts, not the 10:00 shift split.

10. kWh per customer

kWh per customer = kWh in the period / customers in the same period

On the reports, this row is shown only when every device is selected, and it sits above total consumption. It is on both the 6:00 AM table and the midnight table. The customers use the same window as the kWh. It is shown to 2 decimal places. It is kWh per customer, not watts. If there are no customers, the chain board shows a dash.

11. Table occupancy

The business report asks the zone occupancy service for the same dates, the same shift or full day, and the same table. Occupied time is no longer built by grouping camera sightings.

occupied minutes = minutes in the range when at least one person was at that table

peak = highest people count at that table in the range

The occupied-time total on the report is the sum of the tables. A minute when two tables are busy is counted on each table.

People in the restaurant, and tables in use, are the live camera counts. They are shown only while the selected range includes now.

tables in use % = tables with someone now / table count x 100

Entrance zones are not counted as tables.

Minutes before 4 October 2026, 15:57 are estimated from earlier camera tracks. From that moment, the minutes are the live log. The report says so when the selected range starts before that time.

12. Visitor comparisons

change % = (current - previous) / previous x 100

The same zero rules as the home energy card apply. A missing previous period is left blank.

This month's visitors are calendar days from the 1st through today, compared with the same dates last month and the same dates last year. The visitor week is Saturday through today, compared with the same weekdays last week.

Busiest and quietest hour look only at 10:00-23:00, across weekdays, using the average entries for that weekday and hour. A quiet hour with an average of zero is ignored.

Day parts, in Riyadh hours: breakfast 08:00-11:59, lunch 12:00-18:59, dinner 19:00-23:59, late night 00:00-04:59.

13. How numbers are shown

Home energy card: kWh to 1 decimal. Percent to the nearest whole number, with the minus sign kept.

Reports and the chain board: kWh per customer to 2 decimals. Table-use percent to the nearest whole number.

A blank row means the next meter reading does not exist yet, or a slot was missed. It does not mean zero use.